

THE EFFECTS OF POSTACTIVATION POTENTIATION ON SPRINT AND JUMP PERFORMANCE OF MALE ACADEMY SOCCER PLAYERS

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ABSTRACT

Till, KA and Cooke, C. The effects of postactivation potentiation on sprint and jump performance of male academy soccer players. *J Strength Cond Res* 23(7): 1960–1967, 2009—The purpose of this investigation was to evaluate the postactivation potentiation (PAP) effects of both dynamic and isometric maximum voluntary contractions (MVCs) on sprint and jump performance and establish whether PAP methods could be used effectively in warm up protocols for soccer players. Twelve male soccer players performed 4 warm up protocols in a cross-over, randomized, and counterbalanced design. In addition to a control warm up, subjects performed deadlift (5 repetitions at 5 repetitions maximum), tuck jump (5 repetitions), and isometric MVC knee extensions (3 repetitions for 3 s) as PAP treatments in an otherwise identical warm up protocol. After each treatment, the subjects underwent 3 10 m and 20 m sprints 4, 5, and 6 minutes post-warm up and 3 vertical jumps (VJ) at 7, 8, and 9 minutes post-warm up. Repeated measures analysis of variance showed no significant differences in the first 10 m ($p = 0.258$) and 20 m ($p = 0.253$) sprint and VJ ($p = 0.703$) performance and the average 10 m ($p = 0.215$), 20 m ($p = 0.388$), and VJ ($p = 0.529$) performance between conditions. There were also no significant differences in performance responses between the strongest and weakest subjects, but large variations in individual responses were found between the subjects. The findings suggest that there was no significant group PAP effect on sprint and jump performance after dynamic and isometric MVCs compared with a control warm up protocol. However, the large variation in individual responses (-7.1% to $+8.2\%$) suggests PAP should be considered on an individual basis. Factors such as method, volume, load, recovery, and interindividual variability of PAP must be

considered in the practical application of PAP and the rigorous research design of future studies to evaluate the potential for performance enhancement.

KEY WORDS warm up, power, strength, individual responses

INTRODUCTION

Speed, strength, and power are all determinants of athletic performance, and their optimization in training or competition can be enhanced through an appropriate warm up. A warm up is undertaken before any athletic event, with the majority of effects being attributed to temperature-related mechanisms (3). However, a mechanism currently receiving increased research attention is postactivation potentiation (PAP) (2). Postactivation potentiation is defined as an increase in muscle twitch and low-frequency tetanic force after a previous conditioning contractile activity (20). Evidence suggests that PAP may enhance the ability of muscle to produce more force at a faster rate after previous muscle contractions. Over the past decade, research has focused on the effects of PAP on athletic performance using dynamic movements (1,8,13,16,21,22) and isometric maximum voluntary contractions (MVCs) (6,7,9).

Although both dynamic movements and isometric MVCs have been used to elicit a PAP response, a number of exercises and protocols have been used. The majority of research on dynamic exercise in the lower body has used the squat exercise (2,13,16,22), with the number of repetitions, intensities, and rest periods varying among studies. Plyometric exercises have been used by Hilfiker et al. (10) and Masamoto et al. (15) in the form of drop jumps and double-legged tuck jumps, with a number of studies using isometric MVC leg extensions (6,7) to elicit a PAP effect. Therefore, the wide variety of methods used to enhance PAP highlights the uncertainty of the most effective protocol to elicit a PAP response (16), and few studies have been undertaken that compare different methods of eliciting PAP.

The majority of research in the lower body has tested PAP using the vertical jump (VJ) test (8,12,13,21,22), with other studies using the horizontal jump (21) and knee extension performance (7). To date, only 2 studies found (4,16) have examined the effects of PAP on sprint performance.

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Significant improvements in sprint performance were found at 40 m after 3 repetitions at 90% 1 repetition maximum (1RM) of heavy loaded squats (16) and at 10 and 30 m after 10 single repetitions at 90% 1RM of the back squat (4). These studies provide evidence that PAP has a beneficial effect on sprint performance; however, further research is required to support these findings.

Although research on sprint performance is limited, the results obtained from previous studies on jump performance are contradictory. Young et al. (22) and Gourgoulis et al. (8) showed significant improvement in VJ performance, whereas Jensen and Ebben (12), Jones and Lees (13), and Scott and Docherty (21) found no improvement. Gourgoulis et al. (8) examined VJ performance after 5 sets of 2 repetitions at 20%, 40%, 60%, 80%, and 90% 1RM of the half squat exercise and found a 2.39% increase in jump height. An increase of 2.8% in loaded countermovement jump performance also occurred in the study of Young et al. (22) after a set of 5 half squats at 5RM. Jensen and Ebben (12) found a decrease in ground reaction force for countermovement jumps after a set of squats at 5RM, with the jump at 10 seconds being significantly lower than the prejump score. A decrease in countermovement and drop jump height also occurred in 8 males after a set of squats at 5RM at different rest periods (13). Scott and Docherty (21) showed no significant difference in mean and maximal vertical and horizontal jumps after a 5RM back squat. These studies therefore show a large variability in jump performance after a previous muscle contraction to induce PAP. A number of factors such as rest period, number of repetitions, and intensity of lift are methodologic reasons why contradictory results possibly occurred, with factors such as strength, sex, age, and genetics contributing to interindividual variability in response to PAP protocols (19).

This study was designed to assess the uncertainty regarding the most effective method to induce PAP, support the limited evidence on the effects on sprint performance, and provide individual responses to PAP protocols similar to that of Hilfiker et al. (10). The purposes of this study were to a) determine the PAP effects on group sprint and jump performance; b) compare the PAP effects of a weight exercise, plyometric exercise, and isometric MVC on sprint and jump performance; c) determine the effects of PAP on average performance; d) evaluate the effects of strength levels on PAP response; and e) examine the variation in individual responses after PAP protocols. It was hypothesized that sprint and jump performance would significantly improve after the PAP treatments compared with the control warm up.

METHODS

Experimental Approach to the Problem

A repeated measures, cross-over, randomized design involving 4 treatments (control, weight exercise, plyometric exercise, and isometric MVCs) was used to evaluate the effects of PAP on sprint and jump performance and compare different methods of eliciting a PAP response. Sprint times

were measured at 10 and 20 m, and jump performance was assessed using a standard countermovement VJ test (24). The effect of strength on response to PAP treatment was also evaluated alongside the individual responses of each participant.

Subjects

Twelve full-time professional male academy soccer players (age, 18.3 ± 0.72 yr; stature, 176.72 ± 5.03 cm; body mass, 72.1 ± 8.0 kg) participated in the study. All subjects had at least 12 months weight training experience (mean 25.7 ± 6.9 mo), with all players introduced to weight training at the club. Players were in the soccer season when the testing was conducted and competed once a week with resistance training undertaken twice a week. All participants were familiar with the exercises used because they were part of their training program and were also familiar with the 20 m sprint and VJ tests because these were part of their regular fitness testing battery. Institutional ethics approval was obtained, and all subjects gave written informed consent before participating in any of the testing.

Procedures

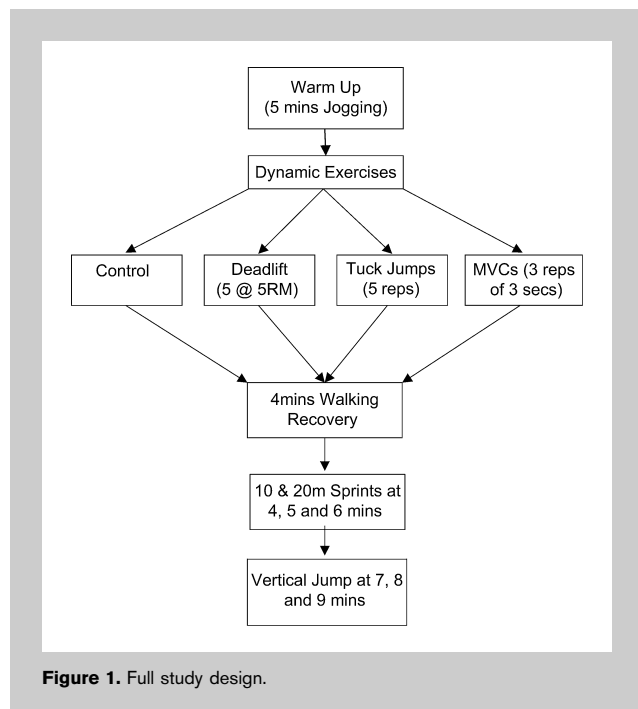
Subjects performed 4 testing sessions over a 4-week period in a cross-over, randomized, and counterbalanced order involving the 3 potentiation protocols (weight exercise, plyometric exercise, MVCs) and the control protocol. The participants performed the testing at the same time of day in an indoor environment with participants instructed not to perform any training the day before testing. Players consumed their normal diet throughout the study but did not drink any caffeinated beverages in the 3 hours before testing. This was controlled through the academy, where players are fed at regular times throughout the day. Before any testing, participants underwent both a strength test to determine their 5RM for the weight exercise and a familiarization session on the isokinetic dynamometer. Figure 1 illustrates the study design.

PAP Protocol

The subjects warmed up by performing 5 minutes of jogging followed by dynamic exercises, followed by 1 of the 4 treatments. After the PAP treatment and before assessment, subjects undertook a 4-minute walking recovery period because performance enhancement after PAP exercise has been reported after a 4-minute recovery period (1,6,16,22). The 4 treatment protocols were as follows.

Control. No PAP treatment was given, and participants had 4 minutes of recovery after the dynamic exercises and were then assessed.

Weight Exercise. Participants performed 5 repetitions of the deadlift exercise at an intensity of 5RM. The deadlift exercise was used because all participants were familiar with the exercise because it was part of their training program, and it had not been previously used in the research. Five repetitions at 5RM was used because Young et al. (22) found performance



improvements using this volume and intensity of the squat exercise.

Plyometric Exercise. Subjects performed 5 maximal repetitions of the double-legged tuck jump exercise. This exercise was used because it is an exercise the subjects used as part of their training programs, uses similar muscles to the deadlift, and was chosen ahead of a high-intensity plyometric exercise (e.g., drop jump) because the subjects had not previously performed this intensity of plyometric exercise. Five repetitions were performed to try to match the volume of the deadlift exercise.

Isometric MVCs. MVCs of the knee extensors were performed on a Cybex 6000 isokinetic dynamometer. The leg was positioned at 90° of knee flexion, and participants were instructed to perform maximal effort leg extensions against the lever arm of the dynamometer. The maximal effort leg extensions were performed for 3 repetitions of 3 seconds per leg with 15 seconds rest in between repetitions. This protocol was used because French et al. (6) found an increase in drop jump and knee extension maximal torque performance after this protocol.

Testing Protocol

After the warm up protocol, 3 20 m sprints at 4, 5, and 6 minutes post-PAP were performed. Sprint times were measured at 10 m and 20 m using Newtest Powertimer 1.0 Testing System timing gates. Subjects began the test 0.5 m behind the initial timing gate in a standing start (17) and were instructed to set off in their own time. After the sprint tests, the subjects performed 3 countermovement jumps on a

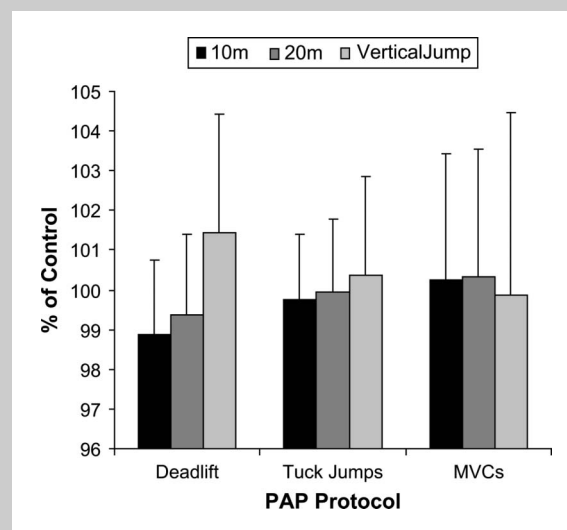


Figure 2. The effect of PAP on performance in 10 and 20 m sprints and vertical jump.

Newtest Powertimer 1.0 Testing System jump mat at 7, 8, and 9 minutes post-PAP treatment. A countermovement jump with no arm movement was performed, which involved subjects starting from an upright position with hands positioned on hips, then flexing the hips and knees, and immediately jumping vertically as high as possible (24).

Statistical Analyses

For statistical calculations, the mean of the 3 sprints and jumps was used. Intraclass correlation coefficients were calculated for the 3 repetitions of sprints and jumps for the control condition using an Excel spreadsheet by Hopkins (11). Intraclass coefficient correlation values were 0.812 to 0.83 for

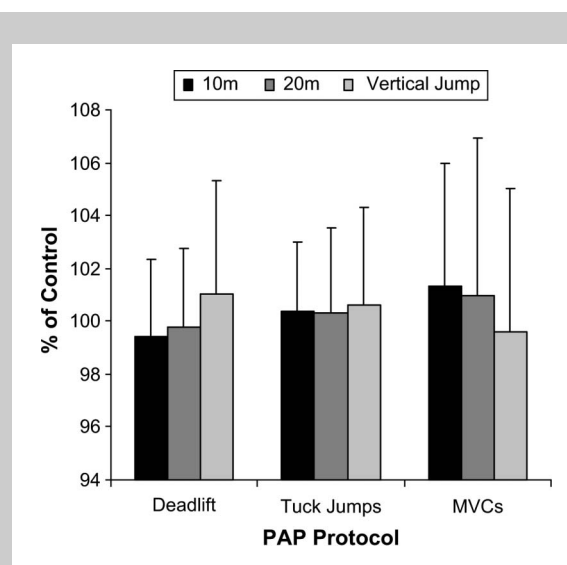


Figure 3. The average effects of PAP on performance in 10 and 20 m sprints and vertical jump over a number of subsequent tests.

TABLE 1. Comparison of strongest versus weakest participants.

		Strongest (<i>n</i> = 6)	Weakest (<i>n</i> = 6)
10 m	5RM dead lift (kg)	72.5 ± 8.22	62.5 ± 8.80
	Relative values (kg)	1.001 ± 0.04	0.874 ± 0.08
	Control	1.791 ± 0.05	1.815 ± 0.07
	Dead lift	1.768 ± 0.04	1.797 ± 0.07
	Tuck Jumps	1.788 ± 0.06	1.808 ± 0.07
	MVCs	1.808 ± 0.07	1.806 ± 0.06
20 m	Control	3.082 ± 0.08	3.121 ± 0.08
	Dead lift	3.067 ± 0.06	3.096 ± 0.08
	Tuck Jumps	3.072 ± 0.08	3.128 ± 0.10
	MVCs	3.104 ± 0.07	3.116 ± 0.10
Vertical jump	Control	40.78 ± 4.71	40.28 ± 5.00
	Deadlift	41.83 ± 4.25	40.28 ± 4.86
	Tuck Jumps	41.33 ± 4.78	40.0 ± 5.08
	MVCs	40.44 ± 4.23	40.50 ± 4.38

*5RM = 5 repetition maximum; MVC = maximum voluntary contraction.

the 10 m sprint, 0.787 to 0.801 for the 20 m sprint, and 0.946 to 0.948 for the VJ. Analysis of results was conducted using a factorial repeated measures analysis of variance test with Bonferroni adjustment on SPSS version 13.0 software (SPSS, Inc., Chicago, IL, USA). The alpha level was set at $p < 0.05$, and the Huynh-Feldt adjustment was used where required based on a test of sphericity. Figures show the results as a percentage of the control protocol performance, with each control performance considered to be 100% of the individual's maximal performance (i.e., a sprint time of less than 100% and a VJ score greater than 100% represent an improved performance). Additional comparisons were also made between the strongest and weakest subjects on the basis of their 5RM deadlift in relation to their body weight. The strongest subjects were those above the average 5RM deadlift in relation to body weight, with the weakest group being those whose value was below the average.

RESULTS

First Sprint and Jump Performance Post-PAP

Figure 2 shows each PAP protocol result relative to the control results for the 10 and 20 m sprint at 4 minutes and VJ at 7 minutes post-PAP. Ten meter and 20 m sprint performance improved to $99.43 \pm 2.93\%$ and $99.79 \pm 2.64\%$ of the control result after the deadlift protocol, with VJ also improving to $101.01 \pm 4.70\%$ and $100.61 \pm 5.92\%$ of the control after the deadlift and tuck jump warm up protocols; however, no significant improvements were found for any protocol.

Average Sprint and Jump Performance

Figure 3 shows the average performance changes for 10 m and 20 m sprint performance at 4, 5, and 6 minutes and VJ

performance at 7, 8, and 9 minutes. Sprint and VJ performance improved compared with the control after both the deadlift (10 m = $98.88 \pm 1.87\%$, 20 m = $99.38 \pm 1.67\%$, VJ = $101.42 \pm 3.18\%$) and tuck jump (10 m = $99.74 \pm 2.01\%$, 20 m = $99.96 \pm 1.84\%$, VJ = $100.37 \pm 3.24\%$) warm up protocols but decreased after the isometric MVCs (10 m = $100.25 \pm 2.99\%$, 20 m = $100.31 \pm 2.47\%$, VJ = $99.85 \pm 4.63\%$). However, there were no significant differences between any of the PAP conditions each assessment.

Effect of Strength Levels

Table 1 shows the results for each test when comparing the 6 strongest (72.5 ± 8.22 kg) and 6 weakest subjects (62.5 ± 8.80 kg) based on their 5RM deadlift. The table illustrates that the strongest group performed better in all tests except for the 10 m sprint and VJ after the MVC protocol. However, there were no significant differences between the strongest and weakest groups in response to the PAP protocols in each assessment.

Individual Responses

Figure 4 shows the mean change in performance of the responders and nonresponders for each PAP protocol and assessment. Figure 4 illustrates the individual changes of each subject for each test and PAP protocol (where no bar appears for a subject, this represents a 0% change). The graph illustrates that the range of responses by each individual varies between participant, test, and PAP method used.

There is a great consistency between the results for the 10 and 20 m sprint, with patterns emerging on the responders and nonresponders to the PAP protocols. For sprint performance, participant numbers 1 and 3 had large positive responses after all PAP protocols, especially the deadlift and MVC protocols, with improvements up to 4.6%. However, there were also participants who responded negatively to the PAP protocols, with participants' 2, 9, and 10 sprint performance decreasing. A 6.4% decrease in 10 m sprint performance was found for participant 9 after the MVC protocol.

Large individual responses were also evident for VJ performance. Participants 9 and 10 responded negatively to all PAP protocols, with only participants 1, 7, and 11 responding positively to all methods. However, large individual gains were made in VJ performance by a number of participants, with participant 1 having the greatest improvement of 8.2%, after the MVC protocol.

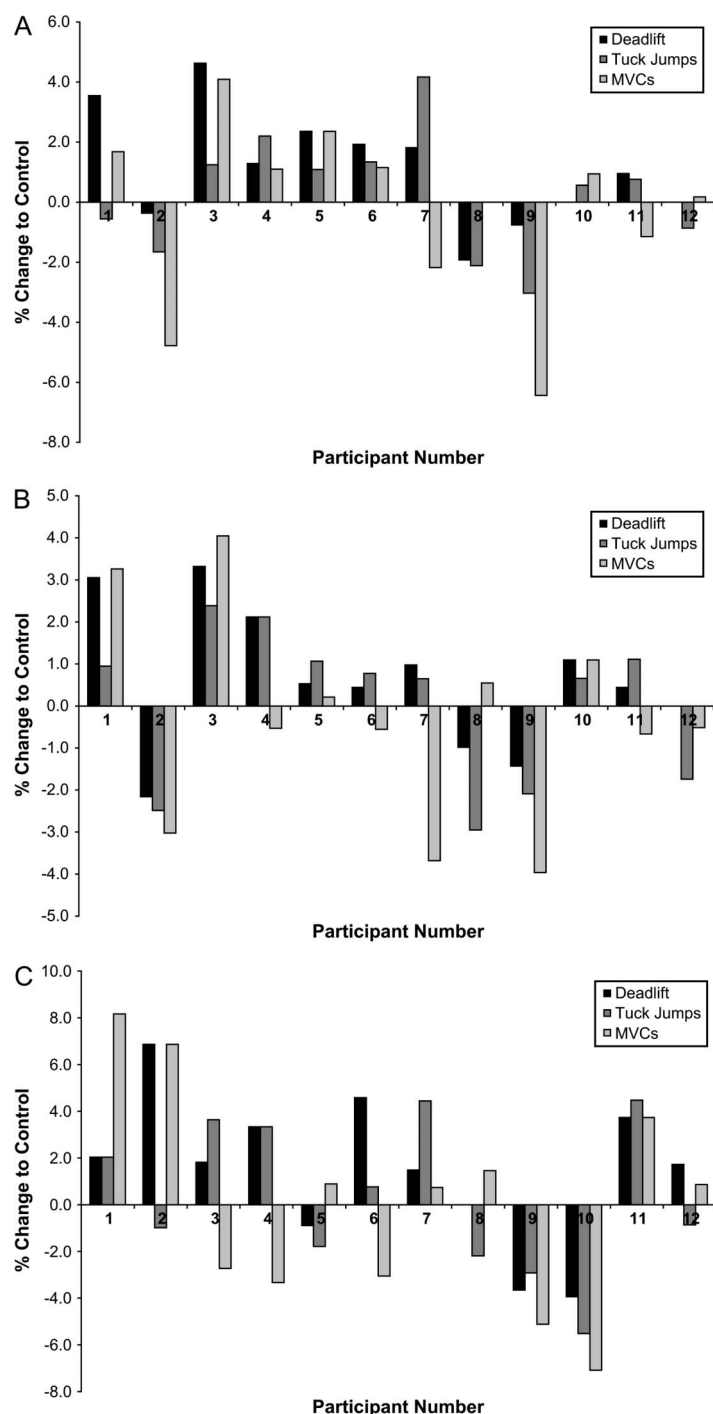


Figure 4. Individual performance changes compared to the control for each PAP method for the (A) 10 m, (B) 20 m and (C) vertical jump assessments.

DISCUSSION

The present study was undertaken to evaluate whether there were any PAP effects of 2 forms of dynamic exercise and isometric MVCs on sprint and jump performance, and it was

hypothesized that sprint and VJ performance would both improve after each PAP warm up protocol compared with the control warm up. The main findings of this study, however, showed no significant group effects of any PAP treatment on sprint and jump performance and that there were no significant differences between any of the PAP methods.

Although no significant changes were evident, 10 and 20 m sprint at 4 minutes post-PAP and VJ performance at 7 minutes post-PAP were improved after the deadlift warm up protocol. Average sprint and VJ performance across the 3 tests were also improved after the deadlift and tuck jump warm ups, showing a positive effect on subsequent performance. However, because no tests at the level of neuromuscular activation (e.g., electromyography or twitch) were undertaken, the mechanism responsible for this trend toward an improved performance could not be assessed. Given the strength of the cross-over, randomized, and counterbalanced research design that was used, we see that the slightly improved performance after the deadlift and tuck jump protocols is attributable to PAP methods used in the warm up protocols, even though the group responses were not statistically significant.

Only 2 studies (4,16) have assessed the effects of PAP on sprint performance. McBride et al. (16) found a significant 0.87% improvement at 40 m ($p = 0.018$) after a set of heavy loaded squats (3 repetitions at 90% 1RM), whereas Chatzopoulos et al. (4) found a 2.6% improvement at 10 m and 1.77% improvement at 30 m 5 minutes after 10 single back squat repetitions. Both studies showed sprint performance could be enhanced after PAP protocols; however, it is difficult to

compare these investigations with the present study because of the different protocols used. A high degree of variability exists in the repeated ballistic action of sprinting (23,24), with participants being unable to perform identical starts and body position changing at the end of a sprint (16) being possible reasons why no significant improvement occurred, especially at short distances such as 10 m and 20 m.

A number of studies have analyzed the effect of PAP response on VJ performance. Studies by Radcliffe and Radcliffe (18), Young, et al. (22) and Gourgoulis et al. (8) found contradictory results to the current study, with jump performance significantly improving by 1.5%, 2.8%, and 2.39%, respectively. A number of studies (12,13,14,21) all found similar results to this study, with no significant improvement found in jump performance; however, these studies did use a variety of vertical, broad, and drop jumps to measure PAP after a set of squats with different rest periods, volumes and intensities of weight exercise, and experience of subjects used compared with the current study. A possible reason why no significant improvement in VJ occurred in this study is that the test was performed 7 minutes after the PAP treatment. Sale (20) stated that the longer the recovery between the end of the conditioning activity and beginning of performance, the greater the recovery from fatigue but also the greater decay of the PAP mechanism. The 7-minute recovery may be too long for PAP to still be evident, with the sprints also causing a greater fatigue than passive or low-intensity exercise. However, because performance was slightly improved after the deadlift and tuck jump protocols, it may be that some PAP was still induced. To find a significant enhancement in VJ, it may have been more appropriate to use a separate testing session from the sprints with a 4-minute recovery period.

A wide variety of methods have previously been used to elicit a PAP response, which highlights the uncertainty of the most effective method to induce a PAP effect (16). This study improved on previous research by comparing 3 different methods to elicit PAP in a cross-over, randomized, and counterbalanced design against a control protocol. Although there was no significant group difference in performance changes after each method, the results did vary, with average performance tending to improve after the deadlift method, improving only slightly after the set of tuck jumps, and decreasing after the isometric MVC protocol. Performance changes may have differed among the 3 methods because of the volume and intensity used for each method. The 5 repetitions at 5RM used for the deadlift was similar to the volume and intensity used by Young et al. (22), Jensen and Ebben (12), Jones and Lees (13), and Scott and Docherty (21) and provided a performance improvement of 1.12%, 0.62%, and 1.42% for 10 m, 20 m, and VJ performance. The use of the tuck jump as a plyometric exercise to induce PAP only improved performance slightly by 0.26%, 0.04%, and 0.37%, respectively, and has only been researched once before by Masamoto et al. (15), who found improved 1RM squat by

0.6% after 3 tuck jumps. However, when Masamoto et al. (15) and Hilfiker et al. (10) used a drop jump, a 3.5% ($p < 0.05$) increase in squat occurred (15), and a 2.2% ($p < 0.05$) improvement in countermovement power occurred (10). It is likely that the trend toward improved performance is caused by similar mechanisms as for the weight exercise, with the explosive-type loading of the plyometric exercise enhancing the excitability of the fast twitch motor units and therefore priming these units to play a more significant role in performance (15). However, 5 repetitions of a relatively low-force intensity plyometric exercise such as the double-legged tuck jump may not have been great enough to create a PAP effect (16). It may have been more appropriate to use a greater number of repetitions or use a higher intensity plyometric exercise (e.g., the drop jump to induce a greater PAP response). The isometric MVCs protocol decreased sprint and jump performance, possibly because of fatigue being caused by the high-intensity contractions separated by short rest periods. The 15-second rest used in this study between the MVC contractions may be the reason performance decreased because of the fatigue of the muscles in between MVCs. Although French et al. (6) found improved drop jump and knee extension performance after 3 repetitions of 3 seconds, they used 3-minute rest periods between contractions. Therefore, the selection of intensity and volume for all exercise types is important in eliciting a PAP response.

An important consideration in the literature is the experience of the athletes used to induce a PAP, with Ebben (5) stating that there is a relationship between strength and PAP. This study demonstrated that stronger subjects generally had faster sprint times and greater VJ performance than the weaker group. However, there was no significant difference in response to PAP between the strongest and weakest groups, although VJ performance did improve in the strongest group after the deadlift by 2.6% and tuck jumps by 1.35% compared with a 0% change and a decrease of 0.6% in the weaker group. These results are similar to those of Gourgoulis et al. (8), who found the strongest group improved VJ by 4.01% compared with only 0.42% in the weakest group. These results support the suggestion that strength levels influence the magnitude of PAP effects, but in the present study, the magnitude of difference in response was not significantly different between the strongest and weakest participants. The differences in speed between the strongest and weakest subjects were also not significant in the study of McBride et al. (16), which may have been a result of there being little difference in strength between the 2 groups.

Other individual factors alongside strength may influence the response to a previous contractile activity, and Gullich and Schmidtbleicher (9) concluded that PAP response varied greatly between individuals, which was apparent in the current study. Performance changes varied between participants, PAP protocol, and assessment, with performance changes ranging from -7.1% to 8.2% compared with the

control protocol. These results suggest that an interindividual variability does exist in response to PAP, with a number of variables including training age, training status, chronological age, genetics (muscle fiber type), sex, and strength levels effecting response to PAP (19). The individual variability, with some participants responding positively and some not responding, has implications for the use of PAP with individual athletes. Coaches and athletes are advised to establish whether they are responders or nonresponders in a training environment before recommending or rejecting PAP protocols as part of a standardized warm up.

The results from this study found that sprint and jump performance did not significantly improve after a weight exercise, plyometric exercise, or isometric MVCs warm up protocol when compared with a control warm up involving no PAP method. There was no group difference between any of the responses in initial or average performance to the PAP methods, and there was also no significant difference in response based on the strength levels of the participants. However, a number of factors need to be considered when evaluating the effects of PAP, including method, volume, load used, recovery time, as well as interindividual variability including training age, training status, chronological age, genetics (muscle fiber type), sex, and strength levels. Therefore, further research is required to clarify the variation in reported effects of different PAP protocols on a number of performance variables, with use of a strong research design such as that used in the present study.

PRACTICAL APPLICATIONS

Although this study failed to show any significant PAP effect on group sprint and jump performance from previous research and there were only small but nonsignificant improvements in this study, it may be possible to enhance individual performance using PAP methods in a warm up protocol. The greatest gains of using PAP appear to be on an individual basis, with a large variability in the individual responses to PAP protocols found in this study. The individual changes in performance varied between a decrease in performance of 7.1% to an improvement in performance of 8.2%. Therefore, coaches, fitness specialists, and players themselves need to examine individual responses to PAP methods during training to establish whether performers are either responders or nonresponders before either implementing or rejecting PAP procedures into individual warm up and performance preparation routines. Given the variability in both individual response and research evidence, practitioners need to consider a number of factors including method, exercises, intensities, volumes, and recovery time to fully benefit from the application of the underpinning theory of PAP. Although it may appear a considerable task to determine individual responses for different athletes, possible improvements of up to 8% would suggest it is worthwhile undertaking to distinguish individuals who do respond positively and consistently to PAP procedures.

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